ISY5002 Pattern Recognition Systems

Traffic Monitoring and Forecasting Workbench: A Modular Prototype

Final Report

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Abstract

Traffic monitoring observes road conditions; short-term forecasting anticipates congestion. This project develops a modular prototype that evaluates the two tasks on their respective data domains. A vehicle-recognition module fine-tunes YOLO26m on MIO-TCD and 320 LTA images with teacher pseudo-labels, and is evaluated on a leakage-aware MIO-TCD split, on LTA teacher agreement, and on a 63-image human-corrected subset of the week-2 validation frames. A per-camera filter removes fixed false positives, and a stacked ensemble of detector counts, background differencing and zero-shot CLIP assigns three congestion levels. A forecasting module compares eight models on PEMSD8 under native and unified nine-channel inputs, with five-seed checks for the leading four. A workbench presents both source-labeled outputs.

On MIO-TCD, fine-tuning raises mAP@0.5 from 0.430 to 0.720. On the human-corrected LTA subset, joint training lowers counting error and raises recall but does not establish an mAP gain (0.478 versus 0.467). On PEMSD8, GWNET has the lowest MAE; MultiHeadSTGCN is competitive but not superior. The filter raises precision from 0.816 to 0.833 at a small recall cost; congestion estimates agree with 74.3% of 400 held-out labels. The prototype does not close the loop from LTA images to local forecasting; calibrated time-aligned series are still required.

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1 Introduction

1.1 Traffic Forecasting in Urban Mobility

Road transport is a major mode of movement in many cities, and the demand placed on urban networks continues to grow with population and economic activity. Congestion imposes costs on travelers, freight operators and the environment, while the physical capacity of the road network is difficult to expand in built-up areas. Operators therefore rely on traffic management to use existing capacity efficiently. Short-term traffic forecasting is a core part of this task: it estimates flow, occupancy or speed over the next 15 to 60 minutes, giving signal controllers, route guidance systems, incident response teams and network planners time to act before congestion forms.

Forecasting methods have developed alongside the sensing infrastructure that supplies them. Loop detectors, radar and probe vehicles provide regular time series at fixed locations, and early methods modeled each location independently. As sensor networks expanded, spatio-temporal graph neural networks became a prominent approach: they represent sensors as nodes in a graph and learn both the spatial correlation between neighboring locations and the temporal dynamics of the series (Yu et al., 2018; Wu, Pan, Long, Jiang, and Zhang, 2019). Benchmarks such as LibCity provide common data splits and implementations, making systematic comparison possible (Wang et al., 2021; Jiang et al., 2023).

Camera networks add a complementary view. Fixed roadside cameras cover wide areas and record individual vehicles, and one-stage detectors can recover vehicle positions and classes from each frame (Ultralytics, 2026). The resulting counts describe what is visible at the camera, while calibrated flow, occupancy and speed series describe the aggregate traffic state used by forecasting models. Combining the two views is a practical route toward monitoring systems that both observe and anticipate traffic.

1.2 Problem Statement and Project Scope

Singapore operates a dense network of fixed roadside cameras that provide continuous visual coverage of its road network. The LTA collection used in this project contains 15,316 images from eight cameras over two weeks, sampled at ten-minute intervals. The images support vehicle recognition and manual monitoring. They do not include calibrated flow, occupancy or speed at the camera locations, and the ten-minute sampling interval gives a discontinuous

view of passing traffic. A forecasting model, by contrast, requires a regular series of aggregate measurements. Public benchmarks provide such series, but for different road networks.

The project addresses this mismatch by building a prototype with three components: a vehicle recognition module, a traffic forecasting module and the Traffic Monitoring and Forecasting Workbench. The recognition component includes the detector, a per-camera false-positive filter and a scene-level congestion estimate; these are post-processing and analysis layers on the same LTA images, not a separate data domain. The detection module is trained on MIO-TCD and jointly fine-tuned on LTA images with teacher pseudo-labels. The forecasting module is trained and evaluated on PEMSD8 under a common protocol that compares eight models. The workbench presents the outputs of both modules and labels their data sources. The design keeps the two data domains explicit and defines the interface through which a calibrated local series would connect them.

The project pursues four questions:

- How well does a detector trained on MIO-TCD transfer to LTA images, measured against teacher labels and against a human-corrected LTA validation subset?
- How do representative forecasting models compare on PEMSD8 under a common protocol, and how does the custom model perform against them?
- How can a workbench present both outputs while keeping their data sources explicit?
- How well do fixed-camera post-processing and visual congestion estimates perform when evaluated against human labels?

The scope is limited to four vehicle classes in still images. Forecasting focuses on traffic flow, with occupancy and speed as secondary analyses. The project is a time-bounded prototype and pipeline design. It establishes the processing chain, evaluation protocol and interface. An operational Singapore forecasting system would additionally require calibrated sensing and longer data collection. Nighttime LTA frames are a particular constraint: low light and small vehicle scale make annotation difficult even for humans.

1.3 Contributions

- An LTA image set of 15,316 frames from eight Singapore cameras over two weeks, with a documented sampling and teacher pseudo-labeling pipeline, and a human-corrected validation subset of 63 week-2 images (1,197 boxes) frozen before evaluation.
- A leakage-aware MIO-TCD evaluation protocol based on scene groups, and a joint fine-

tuning procedure that trains a COCO-pretrained YOLO26m on MIO-TCD and LTA images with teacher pseudo-labels; the model is applied to the full LTA collection.

- A per-camera false-positive filter calibrated by hand for all eight cameras, and a congestion-level estimate for all 15,316 LTA frames that combines the detector, a classical vision method and a zero-shot vision-language model; both are evaluated against human labels, with method selection and testing kept on separate label sets.
- A common-protocol benchmark of eight forecasting models on PEMSD8, covering native and unified nine-channel inputs, with five-seed paired checks for the top four models.
- MultiHeadSTGCN, an STGCN-based model with direct multi-step output, calendar inputs and a selectable spatial operator, examined in a staged ablation and against the benchmark models.
- A web workbench that integrates overview, camera monitoring and forecast views, with historical replay, camera-reliability indicators and a threshold-based risk ranking, together with an integration design for a calibrated local series.

2 Literature Review

Traffic monitoring systems combine sensing and prediction. This section reviews the two lines of work that inform the prototype: vehicle detection in traffic imagery and spatio-temporal traffic forecasting. It also covers the evaluation practices used to compare the resulting models.

2.1 Vehicle Detection in Traffic Imagery

One-stage detectors are the usual choice for traffic imagery because a single forward pass returns boxes and classes at near-real-time speed (Ultralytics, 2026). COCO-pretrained weights are a strong starting point (Lin et al., 2014), but the COCO camera geometry differs from fixed, high-mounted traffic cameras. MIO-TCD provides a traffic-specific benchmark with 137,743 human-labeled localization frames from thousands of traffic cameras (Luo et al., 2018). When target-domain labels are unavailable, pseudo-labeling is a common approach: a teacher model labels unlabeled images, and a student is trained on the resulting hard labels (Lee, 2013). When the same teacher also supplies the evaluation reference, the resulting score measures agreement with the teacher; the LTA results in this report are interpreted on that basis.

2.2 Traffic Forecasting

Spatio-temporal graph models are widely studied for road-network forecasting. STGCN combines temporal convolution with graph convolution (Yu et al., 2018); Graph WaveNet learns an adaptive adjacency and uses dilated temporal convolutions (Wu, Pan, Long, Jiang, and Zhang, 2019); MTGNN learns a graph and a multi-step curriculum (Wu, Pan, Long, Jiang, Chang, et al., 2020); AGCRN uses node-specific embeddings instead of a fixed adjacency (Bai et al., 2020). LibCity provides a common implementation and evaluation environment (Wang et al., 2021; Jiang et al., 2023). In comparative studies, the models are often trained with different input channels, optimizers and early-stopping rules, so an observed difference reflects both architecture and training recipe. The eight models compared in this project span feed-forward, recurrent, fixed-graph, learned-graph and node-embedding designs, which gives the benchmark coverage of the main modeling families rather than a single architecture.

2.3 Evaluation and Benchmarking

Frames from the same camera are highly correlated, so random image-level splits leak near-duplicates into the test set. We group images by scene before splitting and resample whole groups for confidence intervals. Detection accuracy is reported with COCO-style mAP (Lin et al., 2014); uncertainty is estimated with bootstrap resampling (Efron and Tibshirani, 1993). Forecasting comparisons are most informative when every model receives the same input information; the benchmark therefore reports both native configurations and a unified nine-channel setting.

2.4 Synthesis and Research Gap

Across these lines of work, detection and forecasting are usually evaluated on different data representations: annotated images for recognition and calibrated sensor series for forecasting. Detection studies also face a cross-camera adaptation problem when the target cameras differ in viewpoint, scale and lighting from the training benchmark. Forecasting comparisons add a second problem: native configurations differ in input features and training recipes, so a ranking mixes architecture with implementation choices. Monitoring integration raises a third problem: the two components produce different outputs, at different time scales, and their data domains must remain visible to the user. These gaps motivate a modular proto-

type that evaluates each component on its own domain, reports both native and unified-input forecasting comparisons, and makes the boundary between the two data sources explicit in the interface.

3 System Design and Methodology

3.1 System Overview

Figure 1 shows the prototype architecture. The upper path maps LTA camera images to a detection table and to the recognition post-processing layers described in Section 3.3; the lower path maps PEMSD8 time series to forecast metrics. Both outputs feed the Traffic Monitoring and Forecasting Workbench, a web-based interface that presents an overview, a monitoring view and a forecast view. The two paths use separate data domains: LTA camera images for recognition and PEMSD8 road-network series for forecasting; the workbench labels the source of each view. Section 3.7 defines the interface for connecting the modules on a calibrated local series.

3.2 Vehicle Recognition Module

The detection module is trained on MIO-TCD and jointly fine-tuned on LTA camera images. The target-domain set has 320 images: 240 from week 1 for training and 80 from week 2 for validation, stratified by camera and by Singapore-time day/night. A larger COCO-pretrained detector, YOLO26x, generates pseudo-labels for all 320 images at the original 1920-pixel resolution with a confidence threshold of 0.3.

Fine-tuning uses YOLO26m at 736 pixels with batch size 8 for a fixed 30 epochs on an Apple M4 Pro, with one run per recipe. Two recipes are compared: MIO-TCD only, and MIO-TCD augmented with the 240 Singapore training images repeated twice.

All models, including the 80-class COCO baseline, are scored by the same four-class evaluator. The joint model is then run at 1280 pixels over all 15,316 images and writes the CSV tables consumed by the workbench.

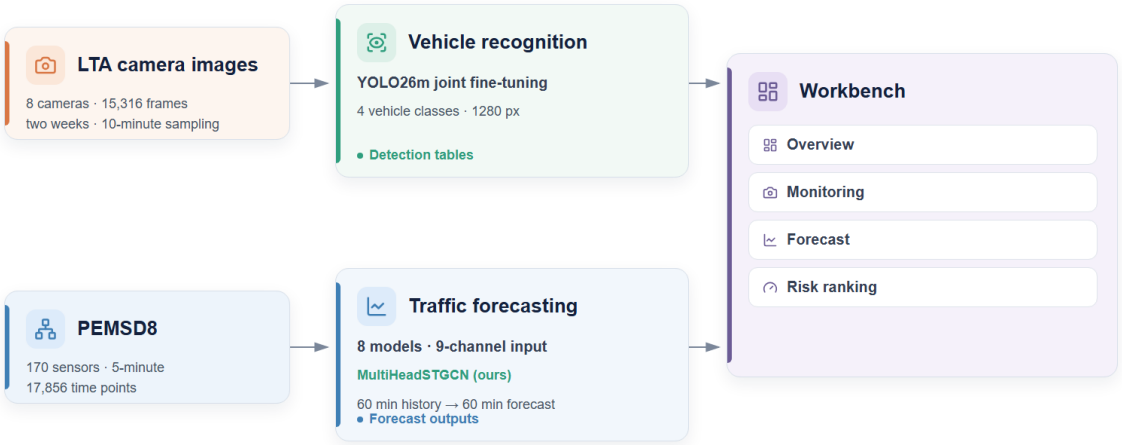


Figure 1: Prototype architecture. The upper path processes LTA camera images for vehicle recognition and its post-processing layers; the lower path processes PEMSD8 time series for traffic forecasting. Both outputs feed the workbench. The camera-filter and congestion layers are described in Section 3.3 and omitted here for clarity.

3.3 Camera Filter and Congestion Estimation

The joint detector runs on fixed cameras, so many of its false positives recur at the same image positions: road signs, traffic lights, lane markings and the central median. A per-camera filter removes them without retraining. Each camera is calibrated once with three elements in normalized image coordinates: a road region, an exclusion set and detection zones. A box is kept only if its bottom-centre point lies inside the road region. It is removed if it matches an exclusion entry, either when its point lies in the entry (point rule) or only when the whole box does (box rule), at all times or only by day or night. The box rule protects vehicles that queue beside a fixed false positive; it is used for a traffic light at camera 4703, where motorcycles wait next to it.

Calibration uses a browser page that plots the bottom-centre points of two weeks of detections for each camera and proposes draft polygons from their density. An annotator adjusts the vertices and accepts or rejects candidate false positives, which are boxes that recur at the same position. The 80 week-2 validation images are excluded from calibration, so the human-corrected images in Section 5.5 were held out from calibration. Because the filter is calibrated per camera, this evaluates image-level correction, not transfer to new cameras. The filter can wrap the detector as a drop-in replacement or filter the existing detection table.

The congestion module rates every LTA frame as free-flowing, moderate or congested, judging only the detection zones: near-field segments that cover the full width of one direction, drawn in the same calibration step. Three voters score each frame independently. Voter A

uses the filtered output of the joint detector: it counts vehicles whose bottom-centre lies in each zone and measures box occupancy, the share of zone area covered by boxes. Voter B is classical image processing: for each camera and hour of day, a per-pixel median over two weeks gives a background, and Canny edges (Canny, 1986) present in the frame but not in the background count as foreground, ignoring over-exposed pixels. The background uses all frames in the collection, including test frames, but no labels; it is an unsupervised, transductive step. Voter C applies a pretrained vision-language model without further training: OpenCLIP ViT-B/16 trained on a LAION-2B subset of LAION-5B (Radford et al., 2021; Cherti et al., 2023; Schuhmann et al., 2022) compares a crop around the zones with three groups of text prompts, and its class probabilities give a congestion score.

When a camera's zones cover two roads, the human label follows the more congested one, so each voter is computed both over all zones and for the busiest zone. Five candidate methods are compared: each voter alone with two cut points, the median of the three voters' levels, and stacking (Wolpert, 1992), a multinomial logistic regression on the log vehicle count, occupancy and the B and C scores. Features, cut points and the primary method are chosen on practice labels only (Section 4.3); the frozen primary method then rates all 15,316 frames.

3.4 Forecasting Module

The forecasting module operates on road-network time series. PEMSD8 provides a calibrated multivariate benchmark with standard evaluation splits: 170 sensors, five-minute observations, and flow, occupancy and speed measurements. The module is trained, tested and compared on PEMSD8, which supports multivariate graph models and comparison with published baselines. The LTA collection provides per-frame vehicle counts at ten-minute intervals; Section 3.7 defines the calibrated series required to connect these counts to the forecasting module.

The module compares eight models under a common protocol: FNN, STGCN, MTGNN, MultiHeadSTGCN, GRU, LSTM, GWNET and AGCRN. The first seven follow their LibCity implementations; MultiHeadSTGCN is the project's custom model. The set covers feed-forward, recurrent, fixed-graph, learned-graph and node-embedding designs. Two input settings are compared: each model's default configuration, and a unified nine-channel setting (flow plus time-of-day plus day-of-week). The unified setting controls the information available to each model, but it does not equalize optimizer, loss function, parameter count or training budget.

The top four models on the unified setting are re-run with seeds 0–4 and compared with paired differences. Every run records its resolved configuration, data provenance, training history and checkpoint selection.

MultiHeadSTGCN is based on STGCN and keeps its temporal and spatial blocks. It changes four aspects. First, it predicts all output steps in one forward pass instead of the LibCity STGCN implementation's single-step output followed by autoregressive rolling. Second, output heads are target-specific: each head produces the complete 12-step horizon, and a single-target run uses one head. The multi-head capacity is exercised only when several targets are trained jointly. The benchmark in this report uses the single-target flow configuration, so its results do not demonstrate a benefit from multiple heads. Third, it adds time-of-day and day-of-week inputs. Fourth, it can use either Chebyshev convolution on the symmetrized PEMSD8 sensor-link adjacency or directed dual-path propagation on the original directed links. The default configuration uses the symmetric Chebyshev variant, a collapsed output head that compresses the temporal axis before the final projection, and the two calendar features. Section 5.11 quantifies each step.

3.5 Workbench

The Traffic Monitoring and Forecasting Workbench is a prototype traffic monitoring and forecasting system. The product concept is a single operator workbench that combines camera monitoring, vehicle counts and short-term congestion forecasts in one interface. The prototype provides three views over the processed outputs:

- **Overview**: a situation summary, a needs-attention queue, camera and forecast health cards, and a camera coverage map.
- **Monitoring**: a camera picker, a time-controlled detection frame, camera reliability metrics (frames processed, zero-detection rate, mean confidence and completeness), and a per-class count timeline.
- **Forecast**: replay-time and horizon controls, a predicted-severity network schematic, an observed and forecast speed curve with an empirical residual band, and a threshold-based risk ranking.

Figure 2 shows the overview: it combines the source labels for both data domains, the needs-attention queue, the module health cards and the camera coverage map. The workbench presents historical replay, and the interface marks this state. It reads the detection ta-

bles and forecast run records, and each view labels its data source. The Monitoring view shows processed frames from the LTA collection; the Forecast view uses PEMSD8 and identifies its sensors as belonging to a public California road network, separate from the LTA cameras. The workbench is implemented as a lightweight web application with static HTML, CSS and JavaScript for the interface and a Python service for data access and consistency checks.

Congestion risk estimation. The Forecast view uses the speed-target run rather than the flow benchmark, because the risk indicator is defined on speed. Detectors are classified as watch below 55 mph, congested below 45 mph and severe below 35 mph. These thresholds are operational choices anchored to a test-period free-flow median of about 65 mph; they are not a traffic-engineering standard. For each detector, the interface takes the worst predicted speed over the selected horizon, including the current observation, and ranks detectors by severity, status and onset. The result is a threshold-based risk indicator intended for comparative visualization, not a validated congestion estimate. The shaded band is the 10th–90th percentile of per-detector residuals on the test period, an empirical error range rather than a calibrated prediction interval.

Camera filter and congestion layers. The Monitoring view also shows two layers that read the outputs of Section 3.3. A camera-filter switch dims the area outside the road region, outlines the detection zones and draws the removed boxes with their reason, and the reliability panel reports the share of boxes removed for the camera. Each frame shows its congestion level, and a weekday-by-hour heatmap gives the share of congested frames for the selected camera together with its test-set accuracy. Without these outputs the view is unchanged. This congestion level describes the LTA camera view; it is separate from the speed-based risk indicator above, which is defined on PEMSD8.

3.6 Implementation and Evaluation Tools

The detection module uses Ultralytics YOLO26 (Ultralytics, 2026) on PyTorch (Paszke et al., 2019), with OpenCV and Pillow for image handling. Fine-tuning and evaluation follow the MIO-TCD protocol (Luo et al., 2018). The project code, workbench, experiment scripts and run records are available in the repository.[1] The forecasting module uses LibCity (Wang et al., 2021;

[1]The code revision used for the reported runs is the `develop` branch: `https://github.com/LOVEIII486/ISY5002-TrafficPrediction/tree/develop`

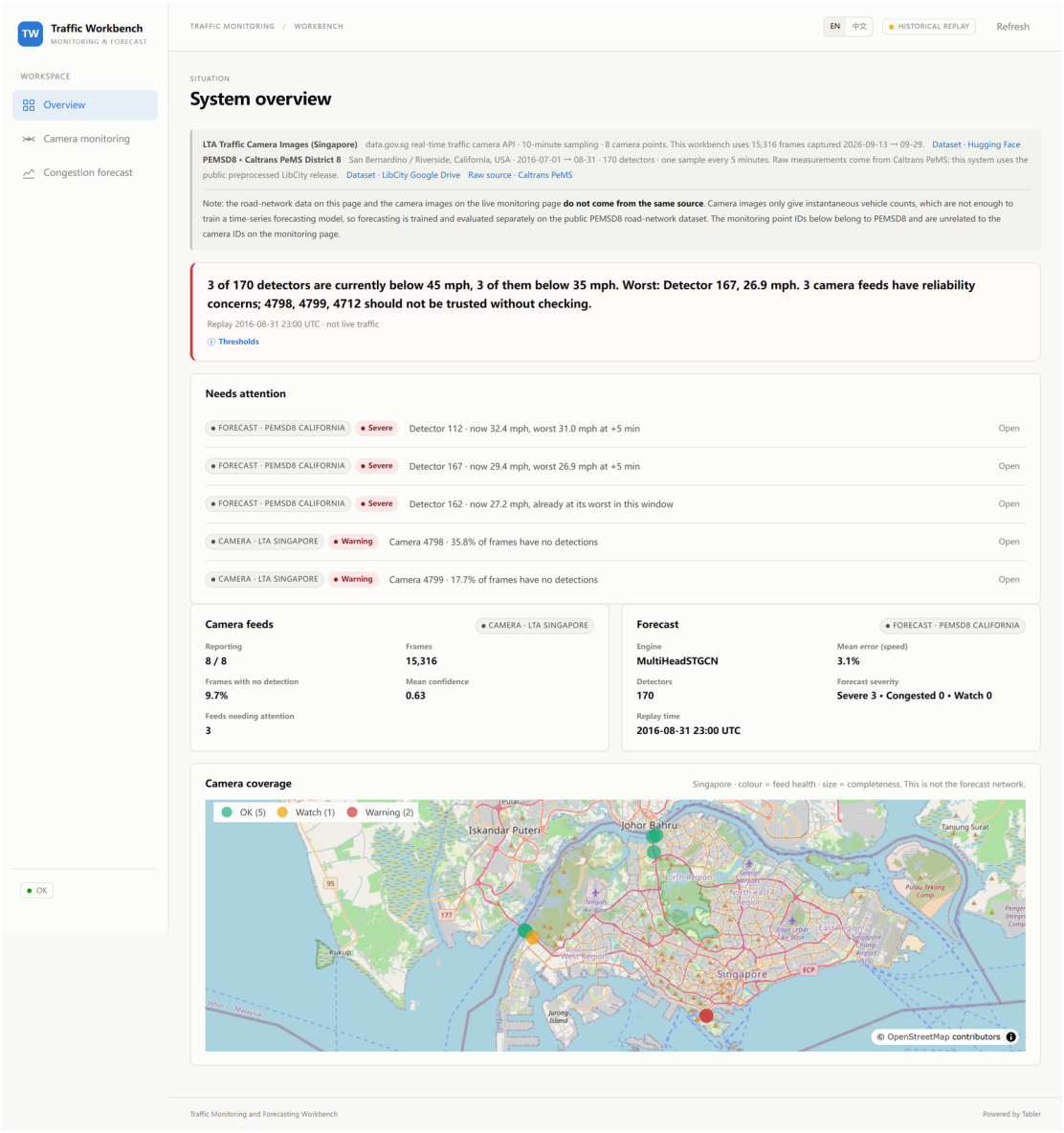


Figure 2: Workbench overview. Each view labels its data source. The page lists forecast and camera items that need attention, summarizes module health, and maps the eight LTA camera feeds.

Jiang et al., 2023), PyTorch, NumPy, pandas and SciPy.[2] The workbench uses Python with static HTML, CSS and JavaScript (Tabler components).[3] The camera filter and the congestion module use OpenCV, scikit-learn and OpenCLIP (Cherti et al., 2023). Large prediction arrays and trained weights are not stored in version control; the benchmark commands regenerate the arrays consumed by the workbench. The human-corrected LTA annotations, manifests and COCO/YOLO exports are released in `datasets/lta_human_v1`; images are referenced by filename and SHA-256 rather than redistributed. Detection accuracy is measured with COCO-style mAP (Lin et al., 2014), and uncertainty is estimated with bootstrap resampling (Efron and

[2] LibCity code: `https://github.com/LibCity/Bigscity-LibCity`
[3] `https://tabler.io/`

Tibshirani, 1993).

3.7 Integration Design

The prototype is organized as three top-level modules with a shared workbench; the camera filter and congestion estimate are recognition post-processing layers, not additional data domains. A fully integrated deployment requires a calibrated, time-aligned traffic series at the camera locations. LTA images provide instantaneous vehicle counts; a forecasting model additionally needs a consistent series of flow, and ideally occupancy and speed, over weeks or months. The integration design has three steps. First, calibrate camera counts against a trusted volume source and aggregate them to fixed intervals. Second, obtain flow, occupancy and speed series for the same locations and align their timestamps with the image feed. Where the existing camera view is too distant or poorly lit, close-range cameras or dedicated traffic sensors can provide clearer measurements. Third, train the forecasting module on the local series and connect both modules to one data domain in the workbench. This design defines the interface between the two modules and the measurements required to complete it.

4 Experimental Setup

4.1 Datasets

The LTA collection contains 15,316 images from eight Singapore cameras over two weeks, sampled at ten-minute intervals.[4] Week 1 and week 2 are disjoint in time. For detection fine-tuning, 240 week-1 images are used for training and 80 week-2 images for validation, stratified by camera and by Singapore-time day/night. The 240 training images also provide the pseudo-labeled target-domain set used in joint fine-tuning. The labels are generated from these images by YOLO26x; they are derived labels generated within the project. Nighttime frames are a particular constraint: vehicles are small and poorly lit, and even human annotation is difficult. Teacher labels and human labels collected at night are therefore the least reliable reference. Figure 3 shows representative frames and labels from both detection datasets.

[4] Collected from the LTA Traffic Images API (data.gov.sg): `https://data.gov.sg/datasets/d_6cdb6b405b25aaaacbaf7689bcc6fae0/view`. The collection is mirrored at `https://huggingface.co/datasets/LOVEIII486/isy5002-lta-traffic-images`.

MIO-TCD provides the human-labeled detection benchmark (Luo et al., 2018).[5] Its localization set contains 137,743 annotated frames; the training release used here contains 110,000 images. We map its eleven classes to four: work vans to car; pickup, single-unit and articulated trucks to truck; bus and motorcycle unchanged. Pedestrian, bicycle and non-motorized-vehicle boxes are dropped, and images containing the class `motorized_vehicle` (vehicles too small to classify) are removed, leaving 89,701 images. Because many images are consecutive frames from the same camera, we link images when they share a file hash, when their 256-bit difference-hash distance is at most 16, when their background thumbnails correlate at 0.95 or above, or when their vehicle-masked background thumbnails correlate at 0.93 or above; connected components form 1,592 scene groups. Whole groups are assigned to pools, and fixed-size samples are drawn from the pools: 8,000 training images, 1,000 development images, 2,000 test images and a 161-image motorcycle diagnostic split. The fixed caps keep the 30-epoch fine-tuning runs tractable and preserve rare-class coverage; the remaining filtered images are not used. The test split contains 31 scene groups and is evaluated once. Our splits are drawn from its training set; the official test labels are not public.

PEMSD8 provides the forecasting benchmark: 170 sensors and 17,856 five-minute observations. The series is distributed by LibCity (Jiang et al., 2023); the raw measurements come from Caltrans PeMS.[6] The common protocol uses a 7:1:2 chronological split, 12 input steps and 12 output steps. The scaler is fitted on the training input windows, and the test window indices are recorded for every run. Figure 4 shows a representative sensor series and its daily profile.

COCO-pretrained weights provide model initialization. The datasets above supply all training and evaluation data. Detection accuracy is measured on the human-labeled MIO-TCD test set; on LTA, results are reported both against teacher-generated labels and against human-corrected labels.

In addition, two human-annotated LTA sets are used. For detection accuracy and the camera filter, an annotator corrected the boxes on the 80 week-2 validation images; 17 images judged too difficult to label reliably were removed, leaving 63 images with 1,197 vehicle boxes. The annotations were initialized from teacher pre-labels and are therefore human-corrected, not

[5]The MIO-TCD localization dataset is available at `https://tcd.miovision.com/`; paper DOI: `https://doi.org/10.1109/TIP.2018.2848705`.

[6]Caltrans PeMS: `https://pems.dot.ca.gov/`; LibCity dataset repository: `https://github.com/LibCity/Bigscity-LibCity-Datasets`.

independent of the teacher. For congestion, 600 frames were labeled free-flowing, moderate or congested in a browser tool that outlines the detection zones and hides the timestamp. The frames were drawn at random per camera within five Singapore-time strata (weekday peak, weekday off-peak, weekend day, evening and late night), never selected by any voter's score, and were split before labeling into 25 practice and 50 test frames per camera. A frame skipped as impossible to judge was replaced by a spare from the same camera and stratum. The practice set contains 107 free-flowing, 47 moderate and 46 congested frames; the test set contains 171, 127 and 102.

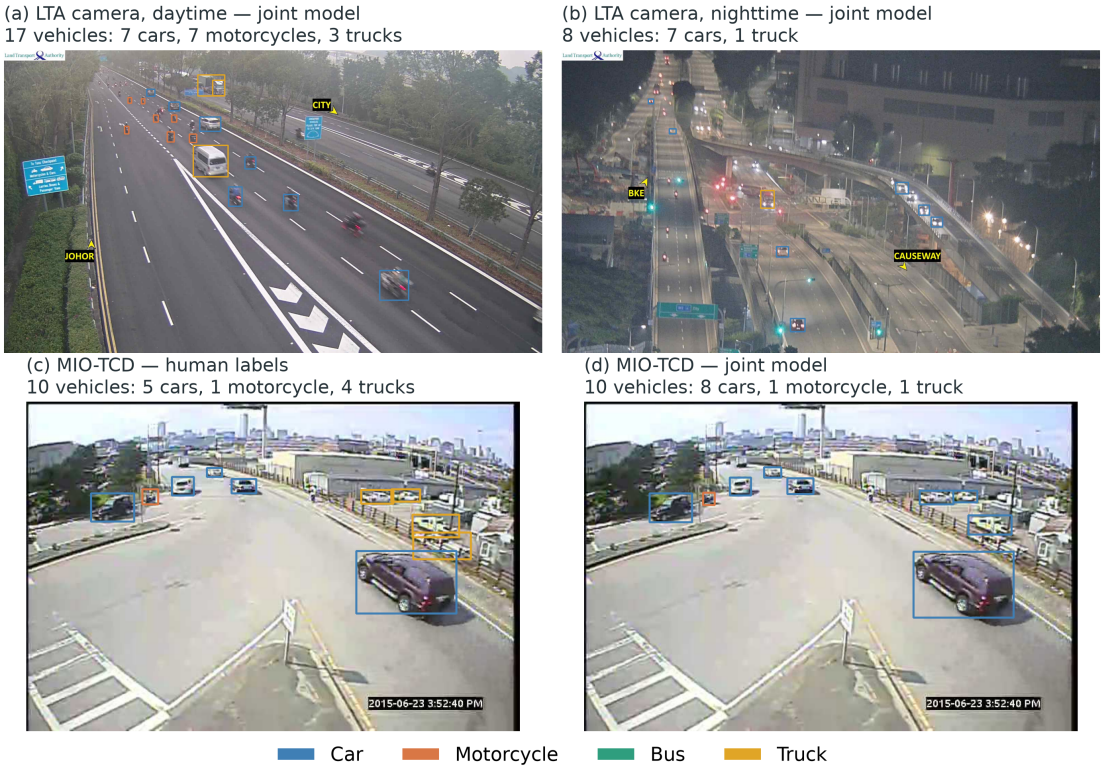


Figure 3: Examples from the detection datasets. (a) LTA daytime and (b) LTA nighttime frames with joint-model detections; (c) a MIO-TCD test image with human labels; (d) the same image with joint-model detections. Box colors denote vehicle classes; titles report detected or labeled counts.

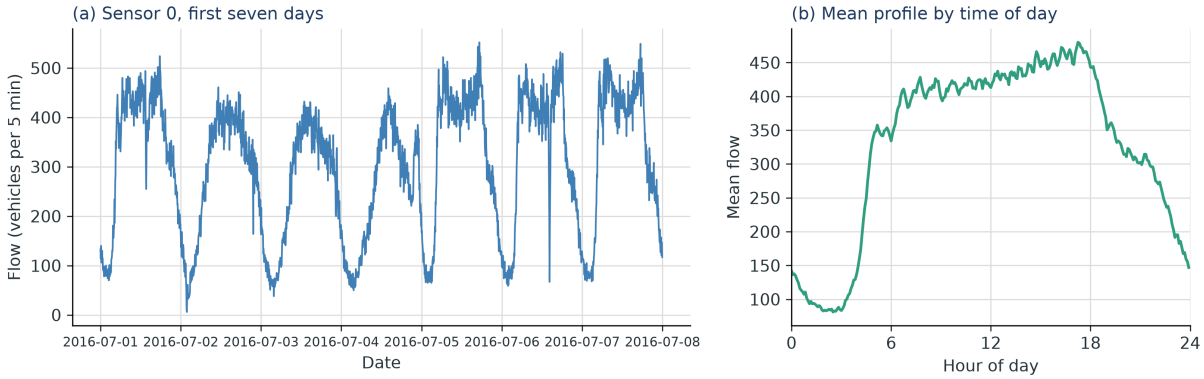


Figure 4: PEMSD8 traffic flow for sensor 0. (a) First seven days of the five-minute series; (b) mean flow by time of day over the full collection. The daily cycle motivates the time-of-day input used in the unified setting.

4.2 Data Domains

The prototype draws on two data sources. LTA images provide camera views and per-frame vehicle counts. PEMSD8 provides calibrated flow, occupancy and speed series for a road network. Each source supports one module: imagery for recognition and calibrated series for forecasting. An end-to-end deployment requires both at the same locations, aligned in time; Section 3.7 defines this interface. The workbench presents the two source-labeled views.

4.3 Evaluation Protocol and Metrics

Detection is scored with COCO-style mAP (Lin et al., 2014) at IoU 0.5 and 0.5:0.95, per-class AP, precision/recall at confidence 0.25, and mean absolute counting error. Confidence intervals come from 2,000 bootstrap resamples of scene groups; paired differences reuse the same resamples. This interval covers scene sampling only, because each detector configuration was trained once.

The camera filter is scored on the 63 human-corrected images by class-agnostic precision and recall at IoU 0.5 and confidence 0.25, before and after filtering, with 2,000 bootstrap resamples of images. These intervals are image-level and do not account for correlation between frames from the same camera, so they are optimistic. Congestion levels are scored by accuracy and linearly weighted Cohen's kappa (Cohen, 1968), which counts confusing free-flowing with congested as twice as severe as confusing adjacent levels, together with precision and recall for the congested level. Features, cut points and the primary method are chosen by five-fold cross-validation repeated 20 times on the practice labels. The primary method is the

candidate with the highest cross-validated kappa; within 0.005 of the best, the simpler candidate is preferred. The test labels are used once, after the configuration is frozen, with 2,000 bootstrap resamples of test frames. These intervals are frame-level; the 50 frames from each camera are correlated, so the intervals are optimistic. Because peak hours are over-sampled, accuracy is also reported with each test frame weighted by its camera and stratum's share of all frames.

Forecasting uses masked MAE, RMSE and MAPE, averaged over the 12 output steps, with 15/30/60-minute breakdowns at output steps 3, 6 and 12. MAE is measured in vehicles per five-minute interval. Masked metrics ignore zero ground truth, with a threshold of 10^{-4}. The scaler is a standard scaler fitted on the training input windows. All models use batch size 64, at most 100 epochs, early stopping with patience 30, and checkpoint selection by validation loss; optimizer, learning rate and schedule follow each model's LibCity default. Runs used PyTorch on Apple MPS with LibCity commit `5a6391d`. Seeds control initialization, shuffling and dropout, while data splits and test windows are fixed. Forecast intervals are paired t-intervals over seeds, not bootstrap intervals; the five-seed results are reported separately.

5 Results and Analysis

5.1 Detection on MIO-TCD

Table 1 reports the frozen test set. Fine-tuning raises mAP@0.5 from 0.430 to 0.709 (MIO-only) and 0.720 (joint). The paired gains over the pretrained detector are 0.279 and 0.291, with intervals that exclude zero. No clear difference was established between the joint and MIO-only models on this dataset (paired difference 0.011, 95% CI −0.024 to 0.048). Counting error roughly halves. Figure 6 gives the per-class breakdown.

Table 1: Frozen MIO-TCD test set (2,000 images, human labels). mAP@0.5 includes 95% scene-group bootstrap intervals; other columns are point estimates.

Model	mAP@0.5	mAP@.5–.95	Car	Moto.	Bus	Truck	Count MAE
Pretrained YOLO26m	0.430 (0.337–0.516)	0.320	0.656	0.136	0.674	0.253	1.212
MIO-TCD only	0.709 (0.626–0.817)	0.561	**0.883**	0.570	**0.920**	**0.463**	**0.650**
Joint fine-tuning	**0.720** (0.645–0.825)	**0.563**	0.874	**0.646**	0.908	0.453	0.698

Note. Bold marks the best value in each column; the shaded row is the joint model. The test set contains 33 motorcycle boxes; the motorcycle diagnostic split provides a larger sample.

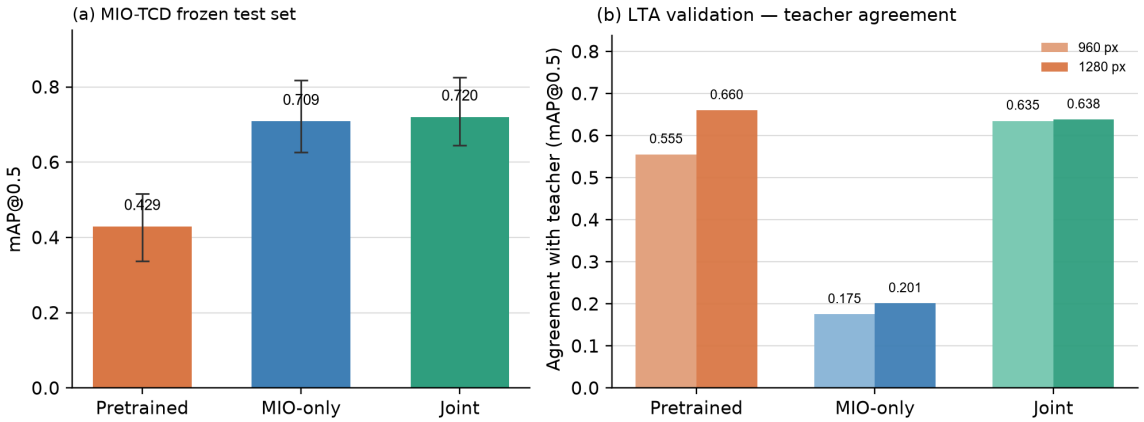


Figure 5: Detection results. (a) Human-labeled MIO-TCD test set; (b) agreement with the YOLO26x teacher on LTA validation. The joint model is shown in green.

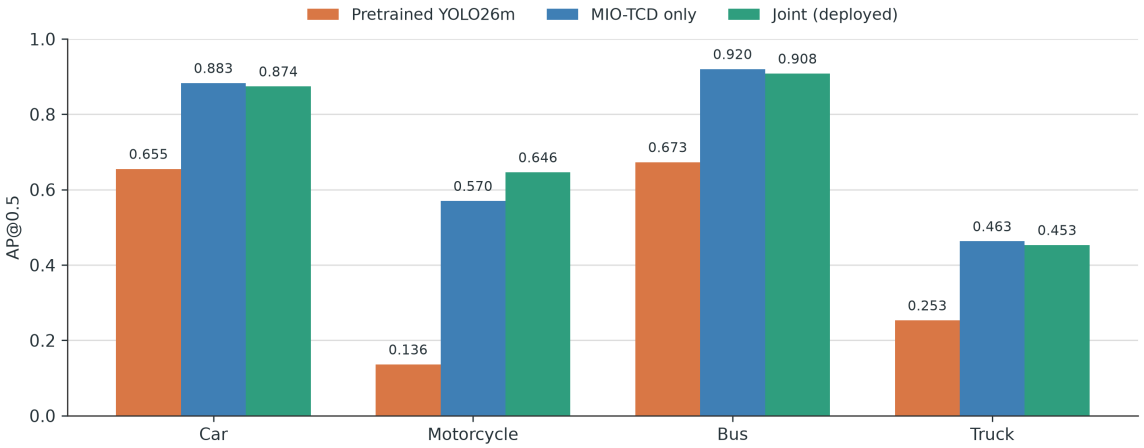


Figure 6: Per-class AP@0.5 on the frozen MIO-TCD test set. The joint model is shown in green and gives the largest gain on motorcycles.

5.2 Cross-Domain Behavior on LTA

The LTA validation set uses the frozen YOLO26x pseudo-labels as its reference, so the figures measure agreement with the teacher. The validation is evaluated at 960 and 1280 pixels: 960 pixels is the lower-resolution setting used during target-domain development, and 1280 pixels is the batch-inference resolution. At 960 pixels the pretrained model reaches 0.555, the MIO-only model 0.175, and the joint model 0.635. The MIO-only result reflects the domain gap between North American close-range views and the Singapore viewpoint.

At 1280 pixels, the batch-inference resolution, the pretrained model reaches 0.660 and the joint model 0.638. Agreement improves substantially over MIO-only training, and the joint model's agreement is close to that of the pretrained model at this resolution. The teacher row is a re-evaluation of YOLO26x at the same inference resolutions after its labels were generated at 1920 pixels; it is a reference point, not a mathematical upper bound on student performance. Nighttime frames are the least reliable part of this proxy: low light and small vehicle scale make both detection and manual annotation difficult. The comparison is therefore a cross-domain proxy; Section 5.3 reports accuracy against human-corrected labels. Figure 7 gives a visual comparison on one frame. On MIO-TCD the ranking reverses: MIO-only training reaches 0.709 mAP@0.5, compared with 0.430 for the pretrained detector (Table 1). The joint model is the compromise between the two domains.

Table 2: LTA validation agreement with the frozen YOLO26x teacher (80 images, week 2).

Model	mAP@0.5 @960	mAP@0.5 @1280	Moto. AP@.5 @960
Pretrained YOLO26m	0.555	**0.660**	0.539
MIO-TCD only	0.175	0.201	0.194
Joint fine-tuning	**0.635**	0.638	**0.670**
Teacher YOLO26x (re-evaluated)	0.707	0.775	0.714

Note. Bold marks the best evaluated model in each column; the shaded row is the joint model. Teacher labels are model predictions. The teacher row re-evaluates YOLO26x at the same inference resolutions used for the students and serves as a reference, not an upper bound.

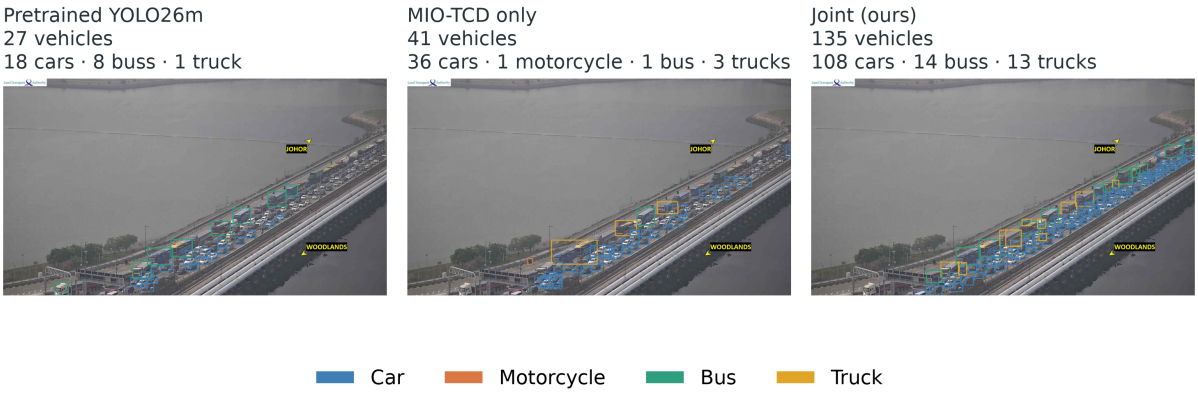


Figure 7: Detection comparison on an LTA daytime frame (camera 2701, input 1280, confidence 0.25). The pretrained detector finds 27 vehicles, MIO-TCD-only training 41, and joint fine-tuning 135. Box colors denote vehicle classes. The counts are model detections and illustrate detection density on one frame; they are not an accuracy measure. Tables 2 and 3 report the quantitative comparison on LTA; on MIO-TCD the ranking reverses (Table 1).

5.3 Human-Corrected LTA Evaluation

The LTA results in Section 5.2 use the frozen YOLO26x teacher as reference. The human-corrected set described in Section 4.1 is a 63-image subset of the 80 week-2 validation images. It is not an independent test set: the same images were used for the teacher-reference comparisons and for the resolution comparison, and the annotations were initialized from teacher boxes. It therefore provides a human-corrected validation measurement, not an independent estimate of accuracy on unseen Singapore cameras.

The subset contains 63 week-2 images from the eight cameras (36 day, 27 night) with 1,197 vehicle boxes: 611 cars, 459 motorcycles, 97 trucks and 30 buses. The annotator started from the teacher boxes, retained 96% of them and added 566, of which 385 were drawn by hand, mostly distant motorcycles. The human-corrected labels were frozen with image-list and label digests before any model was scored against them.

Table 3 reports the three students and the teacher on this set.

On the human-corrected labels the joint model reaches 0.478 mAP@0.5, against 0.467 for the pretrained detector. The paired difference of 0.011 (95% CI −0.043 to 0.060) does not establish an improvement; the two models are also close in the teacher-reference comparison at 1280 pixels (Table 2). Absolute scores are lower than against the teacher (0.478 versus 0.638 for the joint model), because the human-corrected labels add many small vehicles that the teacher missed. The joint model's advantage is coverage rather than overall precision. With the class-agnostic matching of Table 4, its recall is 0.547 against 0.394 for the pretrained

Table 3: Human-corrected LTA validation subset (63 images, 1,197 boxes, input 1280). mAP@0.5 includes 95% intervals from 2,000 image-level bootstrap resamples; other columns are point estimates.

Model	mAP@0.5	mAP@.5–.95	Car	Moto.	Bus	Truck	Count MAE
Pretrained YOLO26m	0.467 (0.420–0.520)	**0.376**	0.753	0.197	**0.483**	0.434	10.46
MIO-TCD only	0.198 (0.152–0.260)	0.132	0.439	0.087	0.151	0.116	14.37
Joint fine-tuning	**0.478** (0.425–0.538)	0.354	**0.767**	**0.300**	0.366	**0.480**	**6.92**
Teacher YOLO26x	*0.572* (0.509–0.637)	*0.486*	*0.808*	*0.298*	*0.530*	*0.653*	*8.46*

Note. Bold marks the best student in each column; the shaded row is the joint model. The teacher row is a reference and is favoured by construction, because the labels were made by correcting its boxes. The table evaluates it at 1280 pixels (0.572 mAP@0.5); at its 1920-pixel labeling resolution it reaches 0.670. Count MAE is in vehicles per image at confidence 0.25.

detector, at a precision of 0.816 against 0.872. Its counting error falls from 10.46 to 6.92 vehicles per image, and its motorcycle AP rises from 0.197 to 0.300. The pretrained detector remains better on buses and on mAP@0.5:0.95. MIO-only training stays far behind at 0.198, confirming the domain gap seen in Section 5.2. All detectors miss many vehicles: the joint model's class-agnostic recall is 0.622 by day and 0.391 at night, and most misses are distant motorcycles.

Three properties of the set limit these figures. The teacher pre-labels favour the teacher and the students distilled from it. The 17 removed images are the densest and darkest frames, mostly from the causeway camera 2702 and at night, so the scores likely overstate performance in those conditions. The set has a single annotator, and the image-level intervals ignore correlation within the eight cameras, so they are optimistic.

5.4 Batch Inference on the LTA Collection

The joint model was run over all 15,316 images at 1280 pixels with confidence 0.25. It produced 279,476 boxes: 187,928 cars, 41,109 motorcycles, 15,111 buses and 35,328 trucks. The mean is 18.25 vehicles per frame, 27.22 during Singapore daytime and 8.93 at night. A total of 1,478 frames returned zero detections, 1,437 of them at night. These frames are concentrated at night, where low light and small vehicle scale limit both detection and manual verification. Nighttime traffic volumes are also lower, so a zero-detection frame does not by itself prove that vehicles were missed. The count is a detection count and may understate nighttime traffic; the workbench's monitoring view presents it alongside camera reliability indicators.

5.5 Camera Filter on Human-Corrected LTA Images

Over the full collection the filter removes 5,536 of the 279,476 boxes (2.0%): 4,110 outside the road regions and 1,426 in exclusion entries. The share ranges from 0.04% at

camera 2701 to 4.7% at camera 4799. Table 4 compares raw and filtered detections on the 63 human-corrected images. Precision rises from 0.816 to 0.833 (+1.7 points, 95% CI +1.0 to +2.9), and false positives fall from 148 to 130. Recall falls slightly, from 0.547 to 0.543 (−0.4 points, 95% CI −1.0 to −0.1).

Of the 23 boxes removed on these images, 18 are false positives and 5 match a labeled vehicle; the latter are mainly water-filled barriers and cars near the edge of a road polygon. The polygons were not changed after the human-corrected labels were seen. The filter does not improve counting: the per-image count error rises from 6.92 to 7.00 vehicles, because the detector already misses about 45% of labeled vehicles and the filter only removes boxes.

Table 4: Camera filter on 63 human-corrected LTA images (1,197 vehicle boxes). Joint model at confidence 0.25, class-agnostic matching at IoU 0.5; intervals from 2,000 image-level bootstrap resamples.

	Raw	Filtered	Change [95% CI]
Precision	0.816	**0.833**	+1.7 pts [+1.0, +2.9]
Recall	**0.547**	0.543	−0.4 pts [−1.0, −0.1]
False positives	148	**130**	−18
Count MAE (vehicles per image)	**6.92**	7.00	+0.08

Note. Intervals are image-level; frames from the same camera are correlated, so these intervals are optimistic.

5.6 Congestion Levels on the LTA Collection

Table 5 reports the five candidate methods. On the practice labels, stacking had the highest cross-validated kappa (0.780) and accuracy (81.5%), and it was frozen as the primary method. On the 400 test frames it reaches 74.3% accuracy (95% CI 69.8–78.5%) and weighted kappa 0.704, against 42.8% for always predicting free-flowing. For the congested level, precision is 0.88 and recall 0.84. Stacking exceeds voter A alone by 2.0 points, but the interval (−1.8 to +5.5) includes zero. Weighted to the full collection, accuracy is 75.8%.

Most errors are between adjacent levels: 69 of the 127 moderate frames were rated free-flowing, and free-flowing and congested were confused only twice. Moderate frames are 32% of the test labels but 24% of the practice labels, which partly explains the drop from cross-validation. Accuracy per camera ranges from 60% at camera 4799 to 94% at camera 4712, on 50 frames each; it is 77% at night and 73% by day.

Practice-set selection also favored the unmasked CLIP crop (76.7% versus 59.6%) and the plain vehicle count (80.0% versus 78.3%).

Figure 8 shows the share of congested frames by weekday and hour. The Causeway

Table 5: Congestion levels: cross-validation on the 200 practice labels (5-fold, 20 repeats) and the single evaluation on the 400 test labels, with 95% bootstrap intervals. κ is linearly weighted; P/R are precision and recall for the congested level.

Method	Practice acc.	Test acc. [95% CI]	Test κ	Congested P / R
Always free-flowing	–	42.8%	0.000	–
A: detector count	80.0%	72.3% [67.7, 76.5]	0.671	0.86 / 0.78
B: background difference	65.3%	62.3% [57.7, 67.0]	0.527	0.62 / 0.83
C: CLIP zero-shot	76.7%	74.0% [69.8, 78.2]	0.703	0.88 / 0.89
Median vote	81.3%	73.0% [68.8, 77.5]	0.696	0.84 / 0.88
Stacking (primary)	**81.5%**	**74.3%** [69.8, 78.5]	**0.704**	0.88 / 0.84

Note. Bold marks the highest accuracy and κ; the shaded row is the primary method, frozen on the practice labels before the test labels were used. Intervals are frame-level and optimistic because the 50 frames from each camera are correlated.

cameras toward Johor (2701) and toward the BKE (2702) are rated congested in 48% and 41% of frames: 2701 most of all on weekday mornings from 06:00 with a second evening peak, and 2702 at both weekday peaks and throughout Saturday. On the Second Link, camera 4703 shows morning and evening peaks (26%), and 4713 is congested mainly at 17:00–18:00 and on Saturday daytime (22%). Cameras 2704 (5%) and 4712 (2%) are rarely congested, and the two Sentosa Gateway cameras almost never (0.2% and 0%). Each weekday-hour cell rests on two or three days (about 11 frames), so one disrupted day can dominate a cell.

5.7 Workbench

The workbench presents the batch-inference outputs through three views. The Overview view reports eight cameras, 15,316 processed frames, 279,476 detections and the forecasting engine used for the speed target, together with camera and forecast health and a needs-attention queue. The Monitoring view exposes per-camera reliability (zero-detection rate, completeness and mean confidence), a time-controlled detection frame and per-class counts. The Forecast view provides replay-time and 15/30/60-minute horizon controls, a predicted-severity network schematic, an observed and forecast speed curve with an empirical residual band, and the threshold-based risk ranking defined in Section 3.5. The interface marks the historical replay state, and each view labels its data source; the Forecast view identifies PEMSD8 as a public California road-network dataset separate from the LTA cameras. The service layer performs consistency checks against the recorded metrics and reports warnings when output sets are incomplete or inconsistent. The Monitoring view additionally shows the camera-filter overlay, each frame's congestion level and the weekday-by-hour heatmap described in Section 3.5.

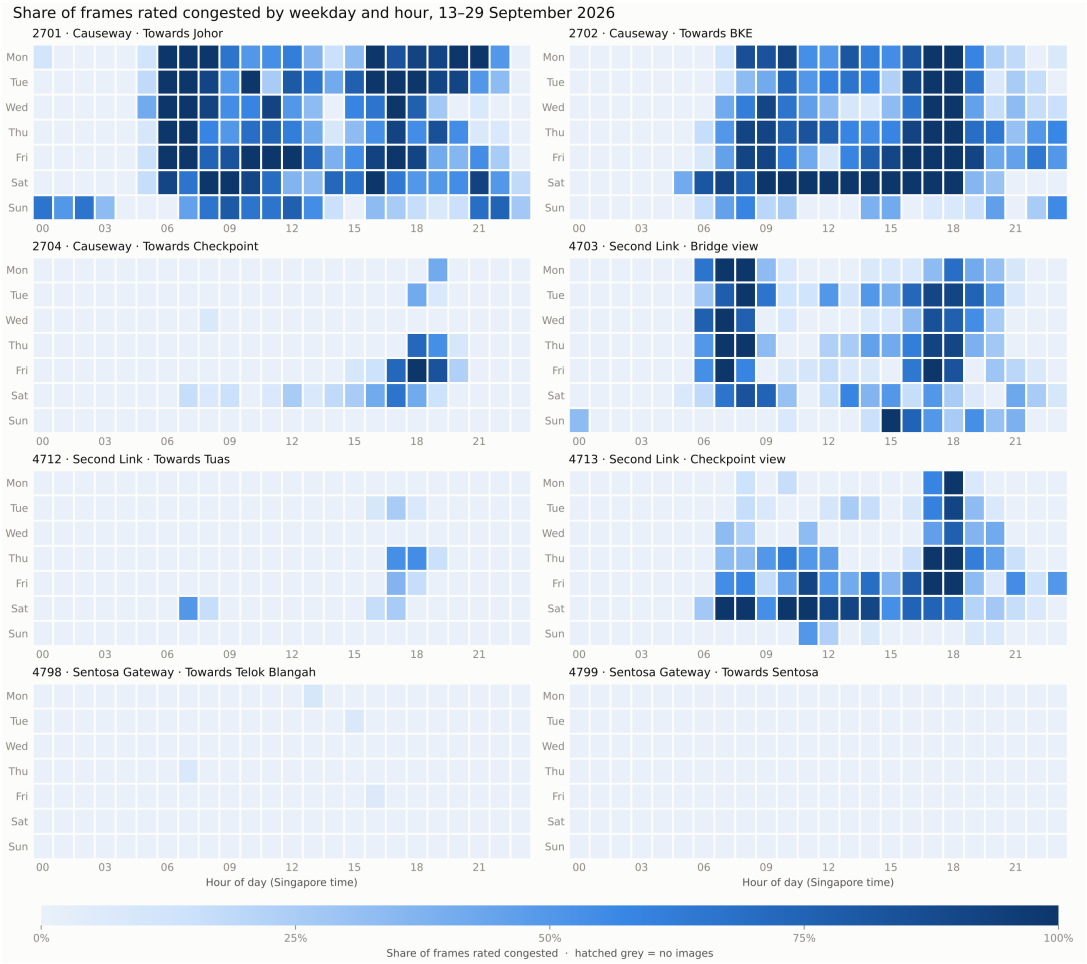


Figure 8: Share of frames rated congested by the primary method, by weekday and hour (Singapore time), 13–29 September 2026. Each panel is one camera; darker cells are congested more often.

5.8 Forecasting: Eight-Model Comparison

Table 6 reports the default-configuration comparison. GWNET has the lowest MAE among the evaluated configurations (14.648), followed by MultiHeadSTGCN (14.839), MTGNN (15.106) and AGCRN (15.612). The configurations differ in input channels and training recipes, so the table compares complete methods under the repository's public protocol. The graph column distinguishes models that use the fixed road adjacency from those that learn a graph or node embeddings.

5.9 The Custom Model: Ablation and Comparison

MultiHeadSTGCN is built on STGCN and keeps its temporal and spatial blocks. The ablation ladder changes one design choice at a time within the custom implementation, but the reference STGCN is the LibCity implementation. The two also differ in training loss (masked MSE for STGCN, masked MAE for the custom model), input features and multi-step strategy.

Table 6: Eight-model forecasting comparison on PEMSD8 traffic flow, default configuration, seed 0.

Model	Graph	Input channels	MAE	RMSE	MAPE (%)
GWNET	Learned graph	1	**14.648**	**23.504**	**9.78**
MTGNN	Learned graph	2	15.106	23.756	10.53
MultiHeadSTGCN	Road adjacency	9	14.839	23.614	9.91
AGCRN	Learned embeddings	1	15.612	24.474	10.31
STGCN	Road adjacency	1	17.436	26.577	10.96
GRU	None	2	19.706	32.363	12.27
FNN	None	1	21.246	32.461	13.30
LSTM	None	2	21.779	36.121	13.24

Note. Bold marks the lowest error in each column; the shaded row is the project's model. Road adjacency is the symmetrized PEMSD8 sensor-link graph; learned graphs and embeddings are estimated from data. Native input channels: 1 = flow; 2 = flow + time-of-day; 9 = flow + time-of-day + seven day-of-week one-hot features. MAE is in vehicles per five-minute interval. Default configurations also differ in training settings; this is a complete-method comparison.

The ladder therefore measures the combined effect of the listed choices, not an isolated architecture effect.

Direct multi-step output lowers MAE from 17.61 to 16.05 (8.9%). Adding time-of-day and day-of-week lowers it to 15.00, an additional 6.5% and a cumulative 14.8% relative to 17.61; day-of-week alone has little effect, while the combination with time-of-day is stronger than either feature alone. Chebyshev convolution on the symmetrized road adjacency lowers MAE to 14.90 (0.7%); this difference is close to the 0.5% variability observed when the same configuration is rerun, so it is an observation rather than a stable gain. The directed dual-path variant gives 14.91. Figure 9 summarizes the ladder.

Three comparisons should be kept separate. In the default-configuration benchmark, STGCN (17.44) and MultiHeadSTGCN (14.84) differ by 14.9%. In the unified nine-channel benchmark, STGCN (16.11) and MultiHeadSTGCN (14.84) differ by 7.9%. The staged ablation above uses a different STGCN reference (17.61). These are complete-recipe comparisons and are not interchangeable.

In the default-configuration table, MAE is 17.44 for STGCN and 14.84 for MultiHeadSTGCN, a 14.9% difference between complete recipes. Under default configurations the model ranks second, after GWNET (14.648) and ahead of MTGNN (15.106) and AGCRN (15.612). Under the unified nine-channel input it ranks fourth, after GWNET (14.365), MTGNN (14.713) and AGCRN (14.777). With five seeds, GWNET's advantage is 4.22% (14.1995 versus 14.8256); the paired differences for MTGNN (14.7443) and AGCRN (14.8913) have confidence intervals that include zero.

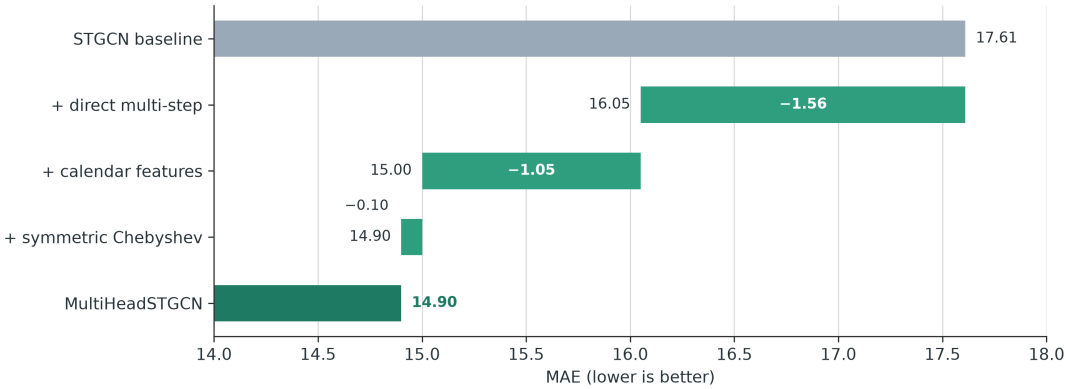


Figure 9: Ablation of MultiHeadSTGCN on PEMSD8 traffic flow. Bars show absolute MAE; labels give the change from the preceding configuration. The x-axis starts at 14 to make the differences visible.

5.10 Unified Input and Multi-Seed Check

Table 7 reports the unified nine-channel setting. At seed 0, GWNET is first (14.365), followed by MTGNN (14.713), AGCRN (14.777) and MultiHeadSTGCN (14.839). The top four were selected by seed-0 test MAE and re-run with seeds 0–4; this selection is exploratory, and the five-seed comparison is descriptive rather than a pre-registered confirmatory test. Their means and paired differences are reported in the same table. GWNET's paired advantage over MultiHeadSTGCN is −4.22% (95% CI −4.98% to −3.47%, better on 5/5 seeds). The paired confidence intervals for MTGNN (−0.55%, CI −1.29% to +0.20%) and AGCRN (+0.44%, CI −0.38% to +1.27%) include zero. Across five paired seeds, no clear performance difference was established between MultiHeadSTGCN and MTGNN or AGCRN. These results suggest comparable observed MAE values, but the limited number of seeds prevents a strong equivalence claim. GWNET achieved the lowest mean MAE among the four models evaluated across five seeds. MultiHeadSTGCN also has the smallest seed-to-seed standard deviation (0.0429). Figure 10 shows the five-seed means and standard deviations, and Table 8 gives the corresponding 15/30/60-minute MAE.

5.11 Ablation Findings

Two ablations are reported. First, calendar features have model-dependent effects. Adding time-of-day and day-of-week reduces MAE by 11.9% for FNN, 7.6% for STGCN, 5.3% for AGCRN, 2.6% for MTGNN and 1.9% for GWNET, and increases MAE by 10.2% for GRU and 18.4% for LSTM. LibCity's RNNs concatenate all nodes and channels into one vector, so the nine-channel input becomes 1,530-dimensional and is harder to optimize with the default

Table 7: Forecasting comparison with unified nine-channel input on PEMSD8 traffic flow. Seed-0 values cover all eight models; mean ± SD and paired differences cover the top four over seeds 0–4.

Model	MAE (seed 0)	MAE mean ± SD (seeds 0–4)	Paired diff vs MultiHeadSTGCN
GWNET	**14.365**	**14.1995 ± 0.1022**	−4.22% ± 0.61%
MTGNN	14.713	14.7443 ± 0.0893	−0.55% ± 0.60%
MultiHeadSTGCN	14.839	14.8256 ± 0.0429	–
AGCRN	14.777	14.8913 ± 0.0773	+0.44% ± 0.67%
STGCN	16.108	–	–
GRU	21.722	–	–
FNN	18.721	–	–
LSTM	25.782	–	–

Note. Bold marks the lowest MAE; the shaded row is the project's model. MAE is in vehicles per five-minute interval. All models use nine channels: flow, time-of-day (one normalized channel) and day-of-week (seven one-hot channels). Mean ± SD describes seed-to-seed variation; the paired differences and their 95% intervals are separate statistics and should not be read as standard deviations. Five-seed statistics cover the top four only. Negative paired differences mean the baseline has lower MAE than MultiHeadSTGCN.

Table 8: MAE by forecast horizon under the unified nine-channel input, mean ± SD over seeds 0–4.

Model	15 min	30 min	60 min	12-step average
GWNET	**13.431 ± 0.084**	**14.219 ± 0.099**	**15.465 ± 0.154**	**14.199 ± 0.102**
MTGNN	13.951 ± 0.090	14.731 ± 0.091	16.035 ± 0.108	14.744 ± 0.089
AGCRN	14.137 ± 0.067	14.893 ± 0.090	16.094 ± 0.121	14.891 ± 0.077
MultiHeadSTGCN	14.233 ± 0.044	14.832 ± 0.045	15.924 ± 0.068	14.826 ± 0.043

Note. MAE is in vehicles per five-minute interval. The 15, 30 and 60 minute horizons correspond to output steps 3, 6 and 12. Values are means over seeds 0–4; ± is one standard deviation across seeds, not a confidence interval.

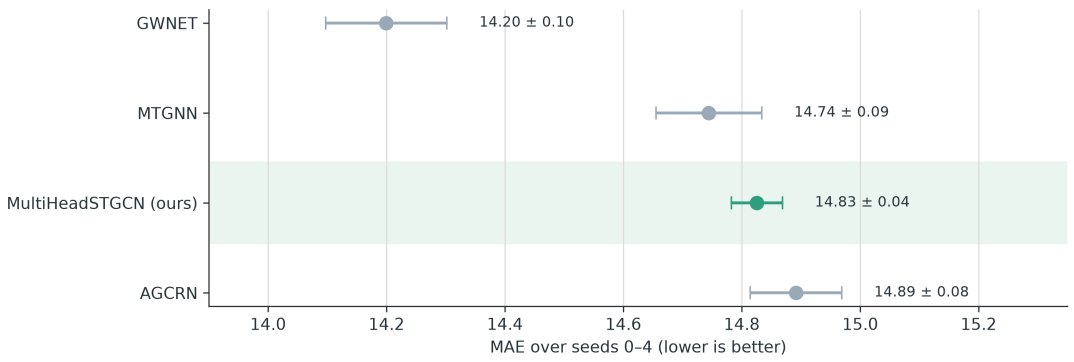


Figure 10: Five-seed MAE on PEMSD8 with unified nine-channel input. Points are means over seeds 0–4; error bars show ±1 SD and are not confidence intervals.

learning rate.

Second, a backbone-matched comparison examines the combined effect of the changes. MultiHeadSTGCN reduces MAE from 17.61 (STGCN) to about 15.00 (14.8%), but because the loss, input features and multi-step strategy also differ, the reduction cannot be attributed to architecture alone. Multi-target training was also evaluated; the multi-target variant, which is the only setting that exercises multiple heads, did not improve flow over the single-target configuration, so the benchmark uses one target head.

6 Discussion

6.1 Detection Performance

On MIO-TCD, where human labels are available, fine-tuning yields a substantial improvement on the evaluated dataset: mAP@0.5 rises from 0.430 to 0.720 and the gain is stable across scene groups. On the LTA images the reference is the YOLO26x teacher, and the joint model reaches 0.638 at the batch-inference resolution, close to the pretrained model's 0.660 and well above MIO-only training at 0.201. Joint training improves agreement on the Singapore validation set while retaining the accuracy gained from MIO-TCD. Against the human-corrected labels on the 63-image validation subset, the joint model matches the pretrained detector in mAP@0.5 (0.478 versus 0.467; the difference does not establish an improvement) but finds more vehicles: recall rises from 0.394 to 0.547 and counting error falls from 10.46 to 6.92 vehicles per image. Its practical benefit on the Singapore cameras is therefore better counting rather than higher overall accuracy, and recall, especially for distant motorcycles and at night, remains the main limitation.

6.2 Camera Filter and Congestion Levels

The camera filter shows that fixed scenes allow a simple, transparent correction of a learned detector: it removes 2.0% of boxes and raises precision by 1.7 points on the human-corrected validation images, without retraining. The gain is small because most errors on LTA frames are missed vehicles, which a filter cannot recover; recall, not precision, limits counting on these cameras.

For congestion, a stacked combination of a trained detector, classical image process-

ing and a zero-shot vision-language model agrees with 74.3% of held-out human labels, and recognizes the congested level with precision 0.88 and recall 0.84. Its 2.0-point gain over the detector count lies within the uncertainty of a 400-frame test, so the result supports the method as a working estimate rather than showing that combining voters is necessary. The main difficulty is the moderate level, which is usually rated free-flowing, the boundary where the visual difference between levels is smallest. Because every design choice was made on practice labels and the test labels were used once, 74.3% is a fair estimate for new frames from the same cameras and period. The levels are a visual judgement of the outlined road, not a calibrated flow or speed, so they do not close the integration gap described in Section 3.7.

6.3 Forecasting Performance

The forecasting module was selected through the common-protocol comparison. GWNET is the best-performing model under the evaluated setting. Across five paired seeds, no clear difference was established between MultiHeadSTGCN and MTGNN or AGCRN, and the limited number of seeds does not support an equivalence claim. The backbone-matched configuration lowers MAE relative to STGCN by about 14.8% in the staged ablation and by 14.9% in the default-configuration benchmark, but these comparisons also change inputs, loss and training strategy; they should not be read as an isolated architectural effect. Under both native and unified inputs, GWNET retains the lowest MAE in this benchmark.

6.4 Module Integration

The detection and forecasting modules are part of one prototype and operate on different data domains. The detection module consumes camera images; the forecasting module consumes road-network time series. The workbench reads both sets of outputs and labels their sources, so a user can inspect the monitoring view and the forecast view with the corresponding data source identified. The separation follows the available data coverage; both modules remain part of the same prototype. Section 3.7 defines the interface for a connected system, and Section 6.5 discusses the implications of moving from the prototype to that system.

6.5 Implications for Future Integration

Section 3.7 defines the data interface; the main implication is that integration is a measurement problem before it is a modeling problem. Camera counts must be calibrated against a trusted volume source and aggregated to a fixed interval, because a detector count per still image is not a flow series. The same locations also need time-aligned flow, and ideally occupancy and speed, over weeks or months. Where the existing view is too distant or poorly lit, close-range cameras or dedicated sensors are the practical way to obtain those measurements.

A second implication concerns model transfer. A forecasting model trained on PEMSD8 learns California sensor dynamics; applying it to a Singapore series would require retraining and a new evaluation on the local split. Detection counts could enter as an exogenous variable, but only after calibration and timestamp alignment. These steps define the work that remains before the workbench can present one local monitoring-and-forecasting workflow rather than two source-labeled views.

6.6 Limitations and Scope

The reported system is a time-bounded prototype and pipeline design. It demonstrates the processing chain and evaluation protocol. An operational LTA-to-Singapore system would require calibrated sensing and longer data collection.

- LTA teacher-agreement results use the YOLO26x pseudo-labels as reference, so the student inherits the teacher's missed motorcycles and false positives. The human-corrected LTA validation subset is small (63 images from eight cameras), was pre-labeled by the teacher and excludes 17 of the densest and darkest frames, so it favours the teacher and likely overstates performance in those conditions.
- Nighttime LTA frames are difficult to annotate even for humans: vehicles are small, poorly lit and often blurred. Teacher labels and human labels collected at night are the least reliable reference.
- LTA agreement depends on inference resolution: the joint model leads at 960 pixels and the pretrained model at 1280 pixels.
- Scene groups provide a content-based proxy for camera identity, so the MIO-TCD test set can retain residual same-camera overlap. In an audit of 40 test images, each paired with its most similar training image, a manual visual review found 9 same-camera pairs.

The test score therefore describes a mix of seen and unseen cameras, not generalization to new cameras.

- MIO-TCD is evaluated on a filtered four-class subset with a project-defined class mapping.
- The workbench risk ranking is a threshold-based indicator defined on predicted speed. It supports comparative inspection but has not been validated against an independent congestion label.
- The human-corrected detection, camera filter and congestion evaluations rely on a single annotator. The congestion test set has 50 frames per camera, and two weeks give each weekday only two or three days, so per-camera and per-slot figures are uncertain.
- Congestion is judged only inside the calibrated detection zones, not along the whole road, and the zones and road regions must be recalibrated if a camera is moved or re-aimed.
- The LTA images are still images at ten-minute intervals. They support vehicle counts per frame but not continuous passage counts at a point. A PEMSD8-style flow series would require continuous video, tracking, lane or region calibration, and validation against a loop or radar sensor.
- Full local integration requires additional IoT measurements (loop detectors, radar, LiDAR or traffic counters) at the camera locations to provide calibrated flow, occupancy and speed. Close-range road cameras are an alternative when the existing camera geometry or lighting is unfavorable.

7 Conclusion

This project developed a modular prototype for traffic monitoring and short-term forecasting, comprising a vehicle recognition module, a traffic forecasting module and a web-based workbench. The system evaluates visual recognition and sensor-based forecasting on their respective data domains while keeping the boundary between those domains explicit.

For vehicle recognition, fine-tuning improved mAP@0.5 from 0.430 to 0.720 on the human-labeled MIO-TCD test subset. Joint training with target-domain pseudo-labels also improved agreement with the teacher model on the LTA validation set relative to MIO-TCD-only training. On the 63-image human-corrected LTA validation subset, the joint model matched the pretrained detector in mAP@0.5 (0.478 versus 0.467) while reducing counting error and raising recall. These findings show the value of traffic-specific adaptation on the benchmark,

while adaptation to the Singapore viewpoint is only partly solved; nighttime frames remain the least reliable part of the evaluation.

Two further components extend the recognition output. A per-camera filter, calibrated by hand, raised precision on the 63-image human-corrected LTA subset from 0.816 to 0.833 at a small cost in recall. A stacked ensemble of the detector count, background differencing and zero-shot CLIP rated all 15,316 frames as free-flowing, moderate or congested, agreeing with 74.3% of 400 held-out human labels. The resulting heatmaps show recurring congestion on the Causeway and Second Link cameras at weekday peaks and on Saturdays.

For traffic forecasting, eight models were evaluated on PEMSD8 under native and unified-input configurations. GWNET achieved the lowest MAE among the evaluated configurations, while MultiHeadSTGCN produced competitive results without a clearly established difference from MTGNN or AGCRN across five seeds. The staged ablation associated direct multi-step prediction and calendar features with substantial reductions in MAE relative to the reference STGCN configuration. Because input features, loss and training strategy also changed, these reductions cannot be attributed to architecture alone.

The workbench presents the monitoring and forecasting outputs in one interface, with source labels, historical replay and a threshold-based risk indicator. The two modules currently operate on different data domains and do not constitute an end-to-end Singapore forecasting pipeline. Future integration requires calibrated, time-aligned local measurements, retraining on the local series and evaluation of the resulting workflow. Overall, the project establishes a modular implementation and an experimental basis for that next stage.

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A Declaration of AI Tool Usage

Table 9 declares all use of AI tools in the preparation of this report, as required by the school's AI policy. All quantitative results reported in this report were produced by executing the project's own code and reading its outputs.

Table 9: Declaration of AI tool usage.

AI Tool	Purpose	How the output was used
Claude Code (Anthropic)	Debugging and optimization of parts of the code; Chinese–English translation	Assisted with implementing and checking code and translating the English document. The authors reviewed and confirmed all content and results.
Codex (OpenAI)	Debugging and optimization of parts of the code	Assisted with implementing and checking code and provided code revision suggestions. The authors reviewed and integrated the suggestions.

Author confirmation: The research design, experiment execution, data interpretation and conclusions in this report were completed by the authors. The AI tools listed above were used for code debugging and optimization and for translation; they did not replace the authors' research judgments. All quantitative results were produced by executing the project's own code and reading its outputs. All external literature cited in the main text is listed in the references.